Evan Reilly

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Technical Skills

Languages: Python, Java, Bash, Javascript, SQL, Go
Technologies: Kubernetes, OpenShift, Git, Docker, Node.js, React, MySQL, MongoDB, Django, Scikit, TensorFlow
Infrastructure: IBM Cloud (IKS & ROKS), Ansible, Jenkins, AWS, EC2, S3, Route53, CloudWatch, GCP, IAM

Experience

IBM Software Engineer — Jan 2022 - Present

- Achieved **10x faster** resource tuning by automating CPU and memory profiling for control plane components
- Built NLP and clustering pipeline to group related customer tickets, improving triage efficiency by **95%**
- Engineered OpenShift patch release pipeline automation, cutting validation time and saving **10+ dev hours per month**
- Developed Kubernetes patch testing automation, reducing patch verification time by **43%**
- Reduced manual ops time by **24+ hours per month** through extending Grafana and Prometheus observability
- Accelerated verification of control plane components by **5x** via automated error detection scripting
- Awarded **Outstanding Technical Achievement** for enabling migration of clusters from OpenShift Toolkit to HyperShift

IBM Software Engineer Intern — May 2021 - Aug 2021

- Automated cluster analysis to accelerate issue resolution, cutting average ticket response time by **56+ hours per month**

Acuity Insurance Software Engineer Intern — May 2020 - Aug 2020

- Developed an intuitive cataloging platform using JavaScript (Dojo) and DB2 to enable efficient analysis of **300K+ policies**
- Migrated mainframe processes to a Pub/Sub architecture, boosting underwriter workflow velocity by **25%**

U.S. Naval Research Laboratory ML Research Intern — May 2019 - Aug 2019

- Developed a StarCraft II reinforcement learning agent in TensorFlow, achieving **15x faster** convergence than baseline models

UW Whitewater Research Assistant — Sep 2018 - Dec 2021

- Published an XAI intentional sequence method for RL agents, producing sequences **364% more informative** than baseline
- Built a CS alumni web app using Java, MySQL, and JSP to visualize statistics for **300+ graduates**
- Delivered technical sessions on database systems to **100+ students**, improving practical understanding
- Analyzed **50 Naval maintenance SOP trajectories** using IRL, optimizing plans for time, cost, and manpower

Projects

Fit to Scale Node.js, Express, React, MySQL, MongoDB, Docker, AWS, EC2, S3, Route53, CloudWatch, Stripe API

- Developed a multi-tenant SaaS platform enabling **10+ trainers** to market and sell services via subscriptions
- Engineered a microservice architecture with API, SQL, and document databases, ingesting **3.5+ million log entries per year**

Bitclout Bot Python, MongoDB, Discord.py

- Developed 10+ features, including borrowing/lending protocol, NFT royalties, role manager, sports betting, and casino games
- Created a Bitcoin-inspired cryptocurrency supporting **100K+ transactions across 2,800+ blocks**

ML for Cardinality Estimation Python, Scikit, MySQL

- Trained ML models for SQL subquery cardinality estimation, reducing median q-error by **30%** versus native estimators

Education

M.S. University of Wisconsin - Whitewater (Computer Science | 4.0 GPA) — Dec 2021

- **Coursework:** Distributed Systems, Machine Learning, Advanced Database Systems, Advanced Algorithm Design & Analysis

B.S. University of Wisconsin - Whitewater (Computer Science, Economics | 3.97 GPA) — May 2020